Impact of Increased Penetration of DFIG-Based Wind Turbine Generators on Transient and Small Signal Stability of Power Systems

Durga Gautam, *Student Member, IEEE*, Vijay Vittal, *Fellow, IEEE*, and Terry Harbour, *Member, IEEE*

***Abstract*—The targeted and current development of wind energy in various countries around the world reveals that wind power is the fastest growing power generation technology. Among the several wind generation technologies, variable speed wind turbines utilizing doubly fed induction generators (DFIGs) are gaining momentum in the power industry. With the increase in penetration of these wind turbines, the power system dominated by synchronous machines will experience a change in dynamics and operational characteristics. Given this assertion, the present paper develops an approach to analyze the impact of increased penetration of DFIG-based wind turbines on transient and small signal stability of a large power system. The primary basis of the method is to convert the DFIG machines into equivalent conventional round rotor synchronous machines and then evaluate the sensitivity of the eigenvalues with respect to inertia. In this regard, modes that are both detrimentally and beneficially affected by the change in inertia are identified. These modes are then excited by appropriate disturbances and the impact of reduced inertia on transient stability performance is also examined. The proposed technique is tested on a large test system representing the Midwestern portion of the U.S. interconnection. The results obtained indicate that the proposed method effectively identifies both detrimental and beneficial impacts of increased DFIG penetration both for transient stability and small signal stability related performance.**

***Index Terms*—Doubly fed induction generator, inertia, sensitivity, small signal stability, transient stability, wind turbine generators.**

I. Introduction

GROWING environmental concerns and attempts to reduce dependency on fossil fuel resources are bringing renewable energy resources to the mainstream of the electric power sector. Among the various renewable resources, wind power is assumed to have the most favorable technical and economical prospects [1]. When deployed in small scale, as was done traditionally, the impact of wind turbine generators (WTGs) on power system stability is minimal. In contrast, when the penetration level increases, the dynamic performance of the power system can be affected.

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D. Gautam and V. Vittal are with the Department of Electrical Engineering, Arizona State University, Tempe, AZ 85287 USA (e-mail: durga.gautam@asu.edu; vijay.vittal@asu.edu).

T. Harbour is a Senior Utility Engineer employed in the utility industry.

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Among the several wind generation technologies, variable speed wind turbines utilizing doubly fed induction generators (DFIGs) are gaining prominence in the power industry. As the performance is largely determined by the converter and the associated controls, a DFIG is an asynchronous generator. Since DFIGs are asynchronous machines, they primarily only have four mechanisms by which they can affect the damping of electromechanical modes (since they themselves do not participate in the modes):

1) displacing synchronous machines thereby affecting the modes;
2) impacting major path flows thereby affecting the synchronizing forces;
3) displacing synchronous machines that have power system stabilizers;
4) DFIG controls interacting with the damping torque on nearby large synchronous generators.

This paper addresses the first two mechanisms listed above. Following a large disturbance, the restoring mechanisms that bring the affected generators back to synchronism, are related to the interaction between the synchronizing forces and the inertia of the generators in the system. In the case of a DFIG, however, the inertia of the turbine is effectively decoupled from the system. The power electronic converter at the heart of the DFIG controls the performance and acts as an interface between the machine and the grid. With conventional control, rotor currents are always controlled to extract maximum energy from the wind. Hence, with the increased penetration of DFIG based wind farms, the effective inertia of the system will be reduced and system reliability following large disturbances could be significantly affected.

Several research efforts have been devoted to address the challenges raised by the proliferation of wind power. According to [2], the DFIG equipped with four-quadrant ac-to-ac converter increases the transient stability margin of electric grids compared to the fixed speed wind systems based on squirrel-cage generators. Reference [3] advocates that the DFIG equipped with power electronics converters and fault ride through capability will have no adverse effect on the stability of a weak grid. Reference [4] focuses on the operational mode of variable speed wind turbines that could enhance the transient stability of the nearby conventional generators. The effect of wind farms on the modes of oscillation of a two-area, four-generator power system has been analyzed in [5]. Reference [6] advocates that the generator types used in wind turbines do not take part in power system oscillations. Rather, the penetration of wind power will have a damping effect due to reduction in the size of synchronous generators that engage in power system oscillations. Reference [7] advocates that increased wind power penetration is accompanied by congestion at weak interconnection lines that leads to reduced damping.

This paper presents a systematic approach based on eigenvalue sensitivity to examine the effect of penetration of DFIGs on a large system. The approach involves detailed analysis of a fairly large system using commercial grade software. The analysis of the critical modes of oscillation is further extended to observe whether or not the system is transiently stable when the particular mode is excited due to a large disturbance.

The paper is organized in seven sections. Section II presents the characteristics associated with DFIGs followed by underlying modeling concepts. Section III discusses the impacts associated with the penetration of DFIGs on power system stability. Section IV details the approach developed to analyze the impact of increased penetration of DFIGs on small signal and transient stability. Section V analyzes the impact on small signal stability. The analysis of transient stability is presented in Section VI. Conclusions that can be drawn from the analysis are presented in Section VII.

II. Characteristics and Modeling of DFIGs

Several generator types are in use for wind power applications today. The main distinction can be made between fixed speed and variable speed wind generator types. Variable speed wind power generation technology encompasses the operation of wind turbines at optimum power coefficient for a wide wind speed range. The two most widely used variable speed wind generator concepts are the converter driven synchronous generator and the DFIG [8]. The present paper, however, includes only the penetration of the latter type of WTG.

The DFIG is a wound rotor induction generator with a voltage source converter connected to the slip-rings of the rotor. The stator winding is coupled directly to the grid and the rotor winding is connected to the grid via a power electronic converter. References [9] and [10] provide a detailed description of the operation of a DFIG.

For power system stability studies, modeling of a DFIG should be considered for steady state analysis as well as for large disturbance dynamic analysis [9].

A. Modeling for Steady State Analysis

Although the wind turbines are distributed within the wind farm, the bulk power from the latter is connected to the grid at a single substation. Consequently, WTGs within the farm are aggregated into a single unit having an MVA rating equal to the summation of the MVA rating of the individual units. Furthermore, as DFIG units have reactive power capability, the wind farm is modeled in a way similar to the conventional generator for steady state analysis and is represented as a PV bus with appropriate VAr limits [10].

B. Modeling for Dynamic Analysis

The power flow provides the initial conditions for dynamic simulation. For power system dynamic simulations, the wind farm is modeled as a single equivalent machine. A complete model of the wind farm with a large number of wind turbines increases the computational burden. Moreover, the assumption is reasonable when the power system under consideration is large and the purpose is to observe the effect of penetration on the external network rather than within the wind farm. Several components that contribute to the dynamic behavior of a DFIG are outlined as follows and included in the analysis conducted [10]:

- turbine aerodynamics;
- turbine mechanical control (also called pitch control) that controls the mechanical power delivered to the shaft;
- shaft dynamics modeled as a two mass shaft, one mass represents rotor/turbine blades and the second represents the generator;
- generator electrical characteristics—as the rotor side converter drives the rotor current very fast, the rotor flux dynamics is neglected and the model behaves as a controlled current source;
- electrical controls—three controllers are used to provide controls for frequency/active power, voltage/reactive power, and pitch angle/mechanical power;
- protection relay settings.

III. Impacts on Power System Stability

In an interconnected system, the ability to restore equilibrium between electromagnetic torque and mechanical torque is determined by the rotor angle stability of each synchronous machine. With the increased number of wind farms in operation, the system experiences a change in dynamic characteristics. Following a disturbance, the change in electromagnetic torque of the synchronous machine can be characterized by two torque components, namely, the synchronizing torque component and the damping torque component. Insufficient synchronizing torque results in non-oscillatory instability whereas insufficient damping torque results in oscillatory instability. These categories of instability in response to the penetration of wind power are discussed in the following sections.

A. Transient Stability

Transient stability is the ability of a power system to maintain synchronism when subjected to a large disturbance. The severe disturbances include equipment outages, load changes or faults that result in large excursion of generator rotor angles. The resulting system response is influenced by the nonlinear power-angle relationship. The time frame of interest in transient stability studies is usually 3–5 s following the disturbance. The duration may extend up to 10–20 s for a very large system with dominant inter-area swings [11].

In a synchronous machine, if during a disturbance the electrical torque falls below the mechanical torque, the rotor will accelerate causing the increase in rotor speed and angular position of the rotor flux vector. Since the increase in rotor angle results in an increase in the generator load torque, a mechanism exists to increase the generator torque so as to match the turbine torque. In case of DFIGs, generator load disturbances also give rise to variations in the speed and the position of the rotor. However, due to the asynchronous operation involved, the position of the rotor flux vector is not dependent on the physical position of the rotor and the synchronizing torque angle characteristic does not exist [12]. The transient stability of a system with wind turbines also depends on factors such as fault conditions and network parameters. Wind speed, however, is assumed to be constant in transient stability simulation involving wind turbines.

B. Small Signal Stability

The small signal stability problem normally occurs due to insufficient damping torque which results in rotor oscillations of increasing amplitude [13]. The eigenvalues of the system matrix A characterize the stability of the system [13]. The change in design and operating condition of the power system is reflected in the eigenvalues of the system state matrix. The effect of the system parameters on the overall system dynamics can be examined by evaluating the sensitivity of the eigenvalues with respect to variations in system parameters [14]. The variable speed WTG design consisting of power electronics converter imparts significant effect on the dynamic performance of the DFIG. The introduction of large amounts of wind generation does have the potential to change the electromechanical damping performance of the system.

IV. Proposed Approach

A. Small Signal Stability

The basis of this study lies on the premise that with the penetration of DFIG based wind farms the effective inertia of the system will be reduced. In this regard, a first step proposed towards studying the system behavior with increased DFIG penetration is to identify how the small signal stability behavior changes with the change in inertia. The approach is thus intended to evaluate eigenvalue sensitivity with respect to generator inertia. The eigenvalue sensitivity with respect to inertia can be expressed as [15]

$$\frac{\partial \lambda_i}{\partial H_j} = \frac{w_i \frac{\partial A}{\partial H_j} v_i^T}{w_i^T v_i} \tag{1}$$

where

H_j	inertia of the jth conventional synchronous generator;
λ_i	ith eigenvalue;
w_i and v_i	left and right eigenvector corresponding to ith eigenvalue, respectively.

The eigenvalue of the state matrix A and the associated right eigenvector (v_i) and left eigenvector (w_i) are defined as

$$Av_i = \lambda_i v_i \tag{2}$$

$$w_i^T A = \lambda_i w_i^T. \tag{3}$$

The eigenvalue sensitivity had been addressed as early as the 1960s [16]. However, the key to the proposed analysis is to examine the **sensitivity with respect to inertia** and identify which modes are affected in a detrimental fashion and which modes are benefitted by the increased DFIG penetration. The following steps are adopted while evaluating the system response with respect to small disturbances:

- Replace all the **DFIGs with conventional synchronous generators** of the **same MVA rating** which will represent the base case operating scenario for the assessment.
- Perform eigenvalue analysis in the frequency range: 0.1 to 2 Hz and damping ratio below 2.5%.
- Evaluate the sensitivity of the eigenvalues with respect to inertia (H_j) of each wind farm represented as a conventional synchronous machine which is aimed at observing the effect of generator inertia on dynamic performance.
- Perform eigenvalue analysis for the case after introducing the existing as well as planned DFIG wind farms in the system.

B. Transient Stability

The machine rotor angle measured with respect to a synchronously rotating reference is considered as one of the parameters to test stability of the system. The severity of a contingency and the trajectory of a system following a disturbance can be assessed by evaluating the transient stability index (TSI). The TSI is obtained from the transient security assessment tool (TSAT) which calculates the index based on angle margin algorithm as follows [17]:

$$TSI = \frac{360 - \delta_{\max}}{360 + \delta_{\max}} \times 100 \qquad -100 < TSI < 100 \tag{4}$$

where $\delta_{\max}$ is the maximum angle separation of any two generators in the system at the same time in the post-fault response. $TSI > 0$ and $TSI \leq 0$ correspond to stable and unstable conditions, respectively.

The objective of transient stability analysis here is to examine if the modes identified in the small signal stability analysis can be excited by the large disturbance and analyze the transient stability performance. Each of the critical modes obtained in the small signal stability study is scrutinized in time domain.

V. Small Signal Stability Analysis

The simulation results carried out for the two different study objectives, namely, small signal stability and transient stability are detailed. With the objective of observing the system response for small and large disturbances, the same base case operating scenario is considered for small signal stability and transient stability study cases.

The study is carried out on a large system having over 22 000 buses, 3104 generators with a total generation of 580 611 MW. All the modeling details provided within the base set of data are retained and represented in the analysis. Within this large system, the study area with a large increase of wind penetration has a total installed capacity of 4730.91 MW. The analysis performed is based on the information provided with regard to the existing and planned increases in wind penetration. A total of 14 wind farms with a total installed capacity of 1460 MW are modeled as DFIGs. This information is used to set up the change cases from the base case provided. The objective is to systematically analyze the impact on system dynamic performance given a large system and the planned penetration of DFIG wind generation.

The system has transmission voltage levels ranging from 34.5 kV, 69 kV, 161 kV, to 345 kV. WTGs are connected at distribution voltage level of 575 V, 600 V, 690 V, and 25 kV. The DFIG model used throughout the work is based on the model developed in [9]. The study is conducted using the package DSATools developed by Powertech Labs, Inc. This includes power flow and short circuit analysis tool (PSAT), transient security assessment tool (TSAT) and small signal analysis tool (SSAT). All new wind farms are represented using the GE 1.5-MW DFIG model available in TSAT.

A. Scenario Description

Four different cases are analyzed. The description of each case is provided in the following:

- Case A constitutes the case wherein all the existing DFIGs in the study area in the original base case are replaced by conventional round rotor synchronous machines (GENROU) of equivalent MVA rating.
- The original base case provided with existing DFIGs in the system is referred to as Case B.
- Case C constitutes the case wherein the penetration of DFIG based WTGs in the study area is increased by 915 MW. The load in the study area is increased by 2% (predicted load growth) and rest of the generation increase is exported to a designated nearby area.
- Case D constitutes the case wherein the DFIG wind farms with the increased wind penetration are replaced by GENROU of corresponding MVA rating. Thus, in Case D the GENROU machines representing the WTGs are of higher MVA rating than in Case A.

The export of increased power from the study area is aimed at keeping the total system power generation constant. This implies that the power export from the study area to the neighboring area is implemented by the concomitant reduction in the generated power in the neighboring area by the same amount.

B. Sensitivity Analysis With Respect to Inertia

The sensitivity analysis with respect to inertia is only conducted for Case A where all machines in the system are represented by conventional synchronous generators. The sensitivity of a given mode with respect to inertia of each wind farm replaced by a conventional synchronous generator is obtained. The sensitivity is evaluated by computing a pair of modes, one of them with the value of inertia in the base case and the other with a "perturbed" value. When computing the mode in the perturbed case, the inertia is increased by 0.5%. The computation is performed using an available feature in SSAT. This feature allows for the evaluation of sensitivity with respect to a wide range of parameters. In this analysis **inertia was chosen to be the sensitivity parameter**. The analysis inherently accounts for the insertion points of the wind farms in the system.

The analysis is carried out only for Case A, for all modes in a range of frequencies from 0.1 to 2 Hz. As the stability of a mode is determined by the real part of eigenvalue, the sensitivity of the real part is examined and the mode which has the **largest real part sensitivity to change in inertia** is identified. Among the several modes of oscillation analyzed, the result of sensitivity analysis associated with the mode having significant detrimental real part sensitivity, in comparison to the real part of the eigenvalue is shown in Table I.

TABLE I
DOMINANT MODE WITH DETRIMENTAL EFFECT ON DAMPING

Real Part (1/s)	Imaginary Part (rad/s)	Frequency (Hz)	Damping Ratio (%)
-0.0643	3.5177	0.5599	1.83

TABLE II
EIGENVALUE SENSITIVITY CORRESPONDING TO THE DOMINANT MODE WITH DETRIMENTAL EFFECT ON DAMPING

No.	Generator Bus #	Base Value of Inertia (s)	Sensitivity of Real Part ($1/s^2$)
1	32672	2.627	-0.0777
2	32644	5.7334	-0.0355
3	32702	3	-0.0679
4	32723	5.548	-0.0367
5	49045	5.2	-0.0383
6	49050	4.6	-0.0444
7	49075	4.2	-0.0475
8	52001	5.2039	-0.0389
9	55612	3.46	-0.0581
10	55678	4.3	-0.0467
11	55881	4	-0.0506
12	55891	4.418	-0.0466
13	55890	5.43	-0.037
14	55889	5.43	-0.0374

The corresponding sensitivity values for the real part of this mode with respect to each of the 14 wind farm generators replaced by conventional synchronous machines of the same MVA rating are shown in Table II.

The real part sensitivities all having negative values as shown in Table II reveal that with the decrease in inertia at these locations, the eigenvalue will move towards the positive right half plane making the system less stable. Participation factor analysis for this mode shows that altogether 26 machines participate in the mode. Participation factors corresponding to each machine are shown in Fig. 1.

The next step in the analysis is to observe if the penetration of DFIGs has beneficial impact in terms of damping power system oscillations. The sensitivity with respect to inertia is examined for positive real part sensitivity. This identifies the mode where the increased penetration of DFIGs in the system results in shifting the eigenvalues of the system state matrix towards the negative half plane. Among the several modes of oscillation analyzed, the result of sensitivity analysis associated with the

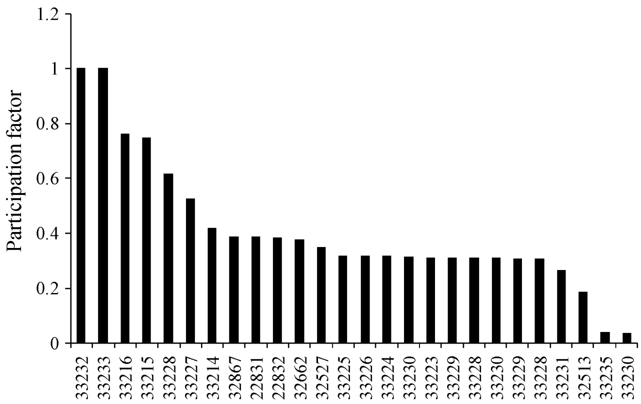

Fig. 1. Participation factor corresponding to the generator speed state for the dominant mode with detrimental effect on damping.

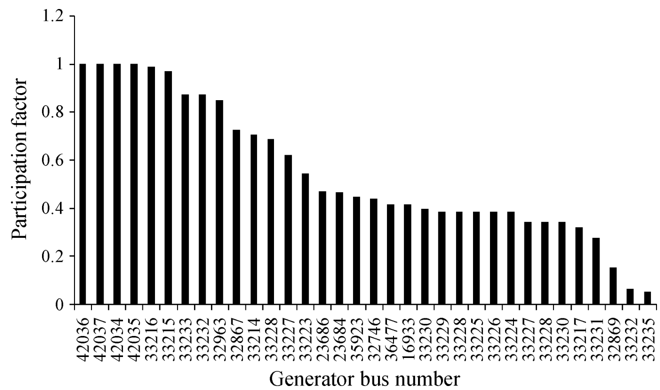

Fig. 2. Participation factor corresponding to the generator speed state for the dominant mode with beneficial effect on damping.

TABLE III
DOMINANT MODE WITH BENEFICIAL EFFECT ON DAMPING

Real Part (1/s)	Imaginary Part (rad/s)	Frequency (Hz)	Damping Ratio (%)
-0.0651	2.8291	0.4503	2.3

TABLE IV
EIGENVALUE SENSITIVITY CORRESPONDING TO THE DOMINANT MODE WITH BENEFICIAL EFFECT ON DAMPING

No.	Generator Bus #	Base Value of Inertia (s)	Sensitivity of Real Part ($1/s^2$)
1	32672	2.627	0.0169
2	32644	5.7334	0.0078
3	32702	3	0.015
4	32723	5.548	0.008
5	49045	5.2	0.0075
6	49050	4.6	0.0092
7	49075	4.2	0.0104
8	52001	5.2039	0.0079
9	55612	3.46	0.0125
10	55678	4.3	0.0098
11	55881	4	0.0107
12	55891	4.418	0.0095
13	55890	5.43	0.0082
14	55889	5.43	0.008

mode having the largest positive real part sensitivity is shown in Table III.

The corresponding real part sensitivity values for each of the 14 wind farm generators represented by conventional synchronous machines are shown in Table IV. The real part sensitivities are all positive in sign indicating that with the decrease in inertia at each of these locations the mode will move further into the left half plane making the system more stable. The participation factor analysis shows that 34 machines participate in this mode. The participation factors corresponding to each machine are shown in Fig. 2.

TABLE V
RESULT SUMMARY FOR CASES A, B, C, AND D FOR DOMINANT MODE WITH DETRIMENTAL EFFECT ON DAMPING

Case	Real	Imaginary	Frequency (Hz)	Damping Ratio (%)	Dominant Machine
A	-0.0643	3.5177	0.5599	1.83	33232
B	-0.0412	3.5516	0.5653	1.16	33232
C	-0.0239	3.5238	0.5608	0.68	33232
D	-0.0427	3.4948	0.5562	1.22	33232

C. Eigenvalue Analysis With DFIG Penetration

Detailed eigenvalue analysis is conducted for Cases A to D in the frequency range of 0.1 to 2 Hz to substantiate the results of the sensitivity analysis.

Table V shows the result of eigenvalue analysis corresponding to the mode listed in Table I which has detrimental impact with increased DFIG penetration. The critical mode is observed in all the four cases. The frequency of the mode is relatively unchanged. It is observed that the damping ratio associated with the mode is reduced from 1.83% in Case A to 1.16% in Case B. The damping ratio has further reduced to 0.68% in the Case C. However, in Case D with DFIGs replaced by conventional machines the damping ratio is improved to 1.22%. The change in damping ratio due the increased penetration of DFIGs accurately reflects the trend indicated by sensitivity analysis. In Cases B and C where the inertia is reduced due to the inclusions of DFIGs the damping has dropped. In Cases A and D where the machines are represented as conventional synchronous machines of equivalent rating the damping is higher. The increased flow into the neighboring area does not affect this mode whereas the change in inertia significantly affects the damping of this mode.

A total number of 28 dominant machines participate in the mode of oscillation in Case A. In Case B, the number of machines participating in the mode increases to 32. In Case C the number of machines participating in the mode further increased to 33. However, the number of machines participating in the mode of oscillation for Case D is 27.

Due to the space limitations the entire list of generators participating in all these modes is not shown here. Nevertheless, this analysis indicates a trend that with the increased penetra-

TABLE VI
RESULT SUMMARY BETWEEN CASES C AND D FOR DOMINANT MODE WITH DETRIMENTAL EFFECT ON DAMPING WITH INCREASED EXPORTS

Case	Real	Imaginary	Frequency (Hz)	Damping Ratio (%)	Dominant Machine
C	-0.0663	3.9097	0.6223	1.7	32963
D	-0.0304	3.7911	0.6034	0.8	32963

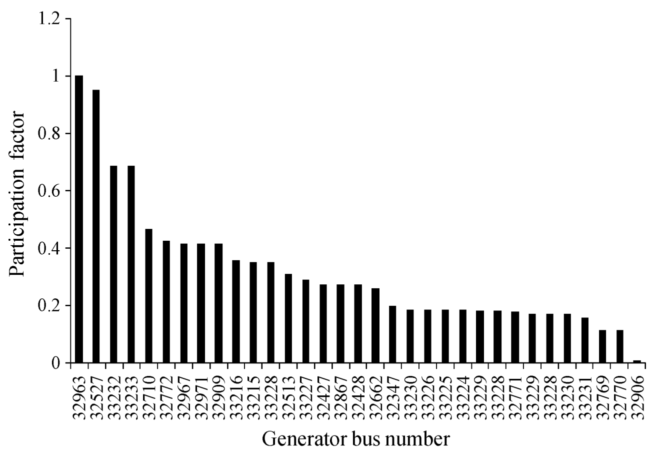

Fig. 3. Participation factor corresponding to the generator speed state for the dominant mode shown in Table VI.

tion of DFIGs in the system, more machines are affected and they participate in the mode of oscillation.

In the course of the modal analysis in the frequency range considered, another mode, which is not observed in Case A and Case B is observed only in Cases C and D and is found to have low damping. The results corresponding to this mode are shown in Table VI. It appears that this mode **manifests largely as a result of the increased exports** to the neighboring area. The machines participating in the mode shown in Table VI with their corresponding participation factor are shown in Fig. 3.

The eigenvalue analysis carried out for the mode with beneficial eigenvalue sensitivity is shown in Table VII. The damping ratio associated with the mode having a beneficial impact has increased from 2.3% in the Case A to 2.55% in the Case B. It is also observed that the damping ratio has further improved to 2.68% in the Case C. However, in Case D with DFIGs replaced by conventional machines the damping ratio is reduced to 2.02%. These results again reflect the trend in damping change identified by the sensitivity analysis and also show the small change in damping as reflected by the sensitivity values in comparison to the mode which was detrimentally affected by the change in inertia.

These results reveal that the eigenvalue sensitivity with respect to inertia provides an effective measure of evaluating the impact of DFIG penetration on the system dynamic performance. The detailed eigenvalue analysis carried out for each of the four cases is found to substantiate the results obtained from sensitivity analysis. The proposed method is found to be valid for identifying both the beneficial as well as the detrimental impact due to increased DFIG penetration.

TABLE VII
RESULT SUMMARY FOR CASES A, B, C, AND D FOR THE DOMINANT MODE WITH BENEFICIAL EFFECT ON DAMPING

Case	Real	Imaginary	Frequency (Hz)	Damping Ratio (%)	Dominant Machine
A	-0.0651	2.8291	0.4503	2.3	42037
B	-0.0725	2.8399	0.452	2.55	33216
C	-0.0756	2.8189	0.4486	2.68	42037
D	-0.0566	2.805	0.4464	2.02	42037

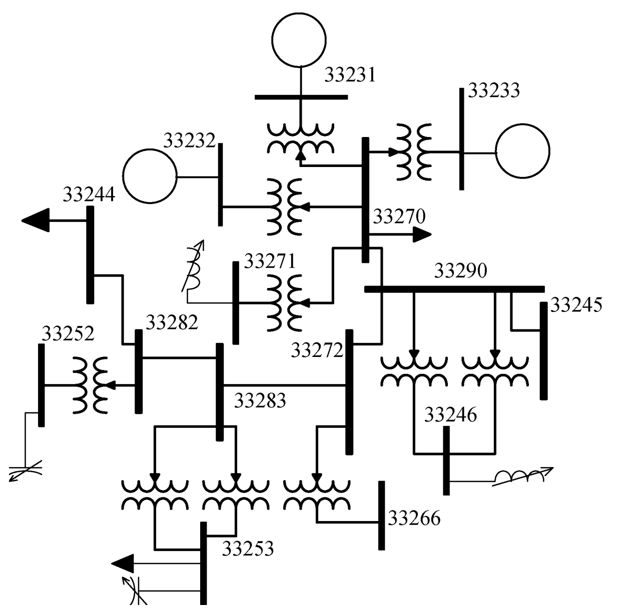

Fig. 4. Single line diagram showing the bus structure near a generator with highest participation.

VI. TRANSIENT STABILITY ANALYSIS

In conducting the transient stability analysis, the objective is to examine if the modes with low damping observed in the small signal analysis could be excited by a large disturbance. In order to identify the disturbance location which will excite the specific mode, the network structure around the machines with the largest participation factors in the specific mode is examined. Simulations are conducted for various three phase faults at several buses followed by outage of the line connected to the faulted bus. The fault is cleared at a suitable time depending on the bus voltage level.

A. Fault Scenario 1—Detrimental Impact on System Performance

The bus structure near the generator 33 232 which has the largest participation factor (see Fig. 1) in the mode detrimentally affected by the penetration of DFIGs (see Table I) is shown in Fig. 4. The objective here is to observe if a large disturbance near generator 33 232 will excite the mode shown in Table I. A three phase fault is applied at bus 33 283 which is a 230-kV bus and is cleared after five cycles. The fault is cleared followed by the clearing of the line connecting the buses 33 283 and 33 282.

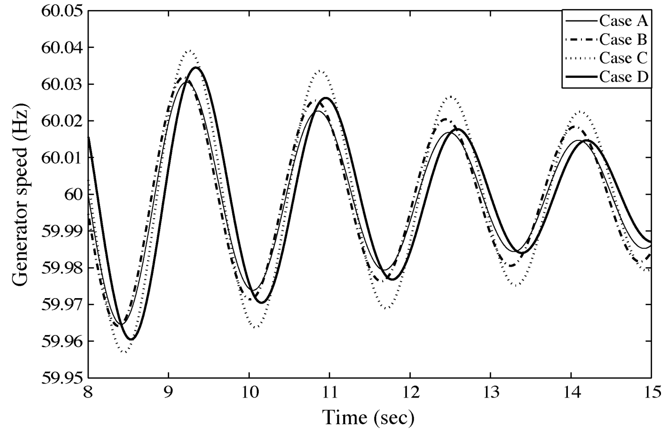

Fig. 5. Bus 32 527 generator speed for Cases A, B, C, and D.

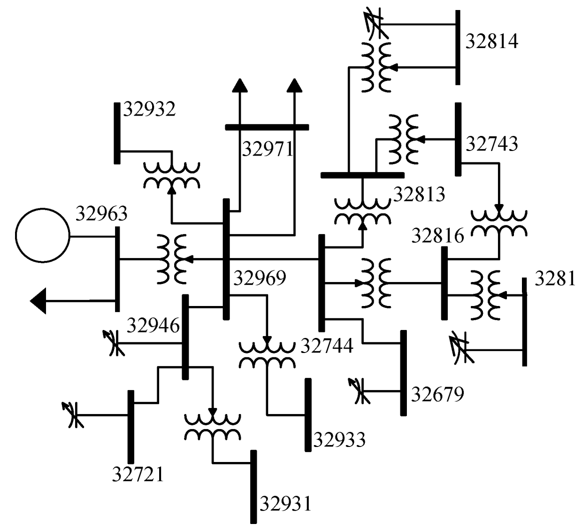

Fig. 6. Single line diagram showing the bus structure near the generator 32 963 with high participation.

As the dominant mode considered from small signal analysis corresponds to the speed state of the machines, the generator speed is observed in time domain. Fig. 5 shows the generator speed corresponding to the generator 32 527 which is one of the machines participating in the mode as shown in Fig. 1. Observing the oscillation corresponding to the last swing of Fig. 5, the least amount of damping is found in Case C which is followed by Case B and Case D. The damping is found to be the highest in Case A. The damping behavior observed in time domain corresponds exactly to the damping ratio provided by the eigenvalue analysis in Table V. This fault scenario is thus found to have detrimental impact on system performance with respect to increased wind penetration.

B. Fault Scenario 2—Examine Low Damping Mode With Increased Export

The bus structure near the generator at bus 32 963 is shown in Fig. 6. This is the generator which has the largest participation factor (see Fig. 3) in the mode given in Table VI. The objective here is to observe if a large disturbance near generator 32 963 will excite the mode shown in Table VI and reflect the result obtained from small signal stability. A three phase fault is applied at bus 32 946 which is at 345 kV. The fault is cleared after 4.5 cycles followed by clearing the line connecting the buses 32 969 and 32 946.

The time domain simulation is carried out to observe the effect of increased DFIG penetration on the system. The relative rotor angle plots for the Cases B and C are shown in Figs. 7–9. The machines represented in these plots are the dominant machines shown in Fig. 3.

The system is found to be transiently secure in Case B whereas it is found to be transiently insecure in Case C. The machines swinging apart in Case C as a result of the fault can be segregated into accelerating and decelerating groups. The machines that accelerate are shown in Fig. 8 and the machines that decelerate are shown in Fig. 9. The result of the disturbance is a large inter area phenomenon as predicted by the small signal analysis shown in Table VI.

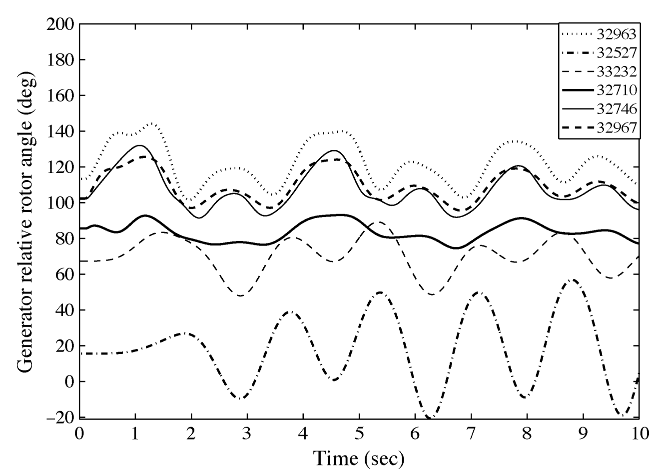

Fig. 7. Generator relative rotor angle for Case B.

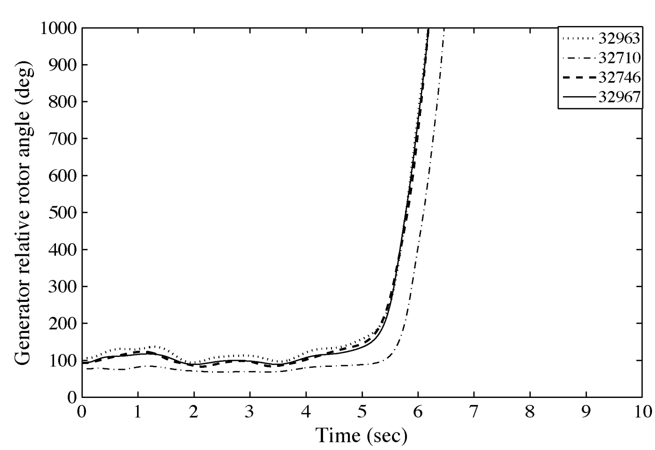

Fig. 8. Generator relative rotor angle for machines accelerating in Case C.

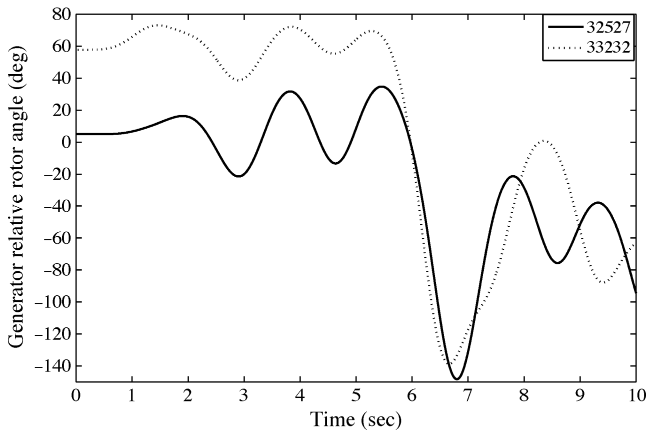

Fig. 9. Generator relative rotor angle for machines decelerating in Case C.

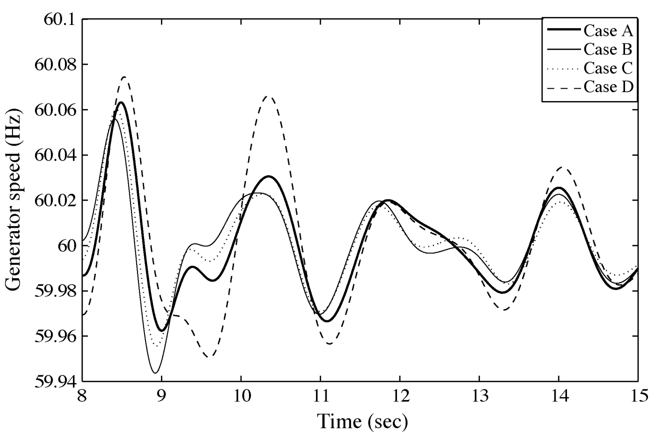

Fig. 10. Bus 33 216 generator speed for Cases A, B, C, and D.

C. Fault Scenario 3—Beneficial Impact on System Performance

Following the same procedure adopted for the other two cases for exciting the required mode following a fault, a three phase fault is applied at a 230-kV bus 33 270 as shown in Fig. 4 and is cleared after 6.6 cycles. The fault clearance is followed by clearing the line connecting the buses 33 270 and 33 290. As the dominant mode considered for small signal analysis corresponds to the speed state of the machines, the generator speeds are observed in time domain.

Fig. 10 shows the generator speed corresponding to the generator 33 216 which is one of the machines participating in the mode as shown in Fig. 2.

The results show that the system is transiently stable for all the four cases and confirms the mode damping ratio results depicted in Table VII. The damping in the plots for Cases A-C are very close to each other and accurately reflect the damping ratio results shown in Table VII. For Case D the damping is markedly lower and verifies that the higher inertia case will have the lower damping. The oscillation damping is increased with the increase in DFIG penetration. This fault scenario is thus found to have a beneficial impact with respect to increased wind penetration. The analysis clearly indicate that the results obtained from sensitivity analysis and from eigenvalue analysis are confirmed by exciting the mode in time domain.

VII. Conclusions

In this paper, the impact of increased penetration of DFIG based WTGs on small signal stability and transient stability is examined for a large system. In order to examine the impact on small signal stability, a systematic approach to pin point the impact of increased penetration of DFIGs on electromechanical modes of oscillation using eigenvalue sensitivity to inertia is developed. In evaluating the sensitivity of specific modes of oscillation with respect to inertia, the DFIGs are replaced by conventional round rotor synchronous machines with the same MVA rating. The sensitivity analysis is performed only for the system where all the generators are synchronous machines. The sensitivity analysis identifies electromechanical modes of oscillations that are detrimentally and beneficially impacted by increased DFIG penetration. The method inherently accounts for the insertion point of the DFIGs in the network. The results of the sensitivity analysis are then confirmed using exact eigenvalue analysis performed by including the DFIGs in the base case and in the increased wind penetration case. Three specific cases of transient stability are also examined. In this analysis the modes observed in small signal analysis are excited by placing specific faults at buses close to the generators having the largest participation factors in the oscillatory modes identified. Transient stability behavior in terms of sufficient system damping and rotor angle stability is analyzed.

For the system operating conditions considered, the analysis conducted indicates that it is possible to identify a certain inter-area mode which is detrimentally affected by the increased DFIG penetration. Moreover, using the concept of participation factors, the specific mode can be excited in time domain.

The system is found to have both beneficial and detrimental impact with the increased penetration of DFIG. Both of these situations observed by sensitivity analysis for small signal stability are also observed in nonlinear time domain analysis by considering corresponding fault scenarios in time domain.

The sensitivity of the real part of the eigenvalue with respect to inertia evaluated for a system where the DFIGs at their planned insertion points in the network are replaced by equivalent round rotor synchronous machines provides a good metric to evaluate the impact due to increased DFIG penetration on system dynamic performance. Both detrimental and beneficial impacts of increased DFIG penetration can be identified. The eigenvalue sensitivity analysis together with the detailed eigenvalue analysis carried out for each of the four cases considered is also substantiated by the results obtained from time domain simulation.

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References

[1] S. Heier, *Grid Integration of Wind Energy Conversion Systems*. Chichester, U.K.: Wiley, 2006.

[2] M. V. A. Nunes, J. A. P. Lopes, H. H. Zurn, U. H. Bezerra, and R. G. Almeida, “Influence of the variable-speed wind generators in transient stability margin of the conventional generators integrated in electrical grids,” *IEEE Trans. Energy Convers.*, vol. 19, no. 4, pp. 692–701, Dec. 2004.
[3] E. Muljadi, C. P. Butterfield, B. Parsons, and A. Ellis, “Effect of variable speed wind turbine generator on stability of a weak grid,” *IEEE Trans. Energy Convers.*, vol. 22, no. 1, pp. 29–35, Mar. 2008.
[4] N. R. Ullah and T. Thiringer, “Effect of operational modes of a wind farm on the transient stability of nearby generators and on power oscillations: A Nordic grid study,” *Wind Energy*, vol. 11, pp. 63–73, Sep. 2007.
[5] J. J. Sanchez Gasca, N. W. Miller, and W. W. Price, “A modal analysis of a two-area system with significant wind power penetration,” in *Proc. 2004 IEEE PES Power Systems Conf. Expo.*.
[6] J. G. Slootweg and W. L. Kling, “The impact of large scale wind power generation on power system oscillations,” *Elect. Power Syst. Res.*, vol. 67, pp. 9–20, 2003.
[7] A. Mendonca and J. A. P. Lopes, “Impact of large scale wind power integration on small signal stability,” in *Proc. 2005 Int. Conf. Future Power Systems*.
[8] M. Poller and S. Achilles, “Aggregated wind park models for analyzing power system dynamics,” in *Proc. 4th Int. Workshop Large-Scale Integration of Wind Power and Transmission Networks for Offshore Wind Farms*, 2003.
[9] GE Energy, Modeling of GE Wind Turbine-Generators for Grid Studies, version 3.4b, Mar. 2005.
[10] CIGRE Technical Brochure on Modeling and Dynamic Behavior of Wind Generation as it Relates to Power System Control and Dynamic Performance, Working Group 01, Advisory Group 6, Study Committee C4, Draft Rep., Aug. 2006.
[11] P. Kundur, J. Paserba, V. Ajjarapu, G. Andersson, A. Bose, C. Canizares, N. Hatziargyriou, D. Hill, A. Stankovic, C. Taylor, T. V. Custem, and V. Vittal, “Definition and classification of power system stability,” *IEEE Trans. Power Syst.*, vol. 19, no. 2, pp. 1387–1401, May 2004.
[12] F. M. Hughes, O. A. Lara, N. Jenkins, and G. Strbac, “Control of DFIG-based wind generation for power network support,” *IEEE Trans. Power Syst.*, vol. 20, no. 4, pp. 1958–1966, Nov. 2005.
[13] P. Kundur, *Power System Stability and Control*. New York: McGraw-Hill, 1994.
[14] P. J. Nolan, N. K. Sinha, and R. T. H. Alden, “Eigenvalue sensitivity of power systems including network and shaft dynamics,” *IEEE Trans. Power App. Syst.*, vol. PAS-95, pp. 1318–1324, 1976.
[15] T. Smed, “Feasible eigenvalue sensitivity for large power systems,” *IEEE Trans. Power Syst.*, vol. 8, no. 2, pp. 555–561, May 1993.
[16] J. E. Van Ness, J. M. Boyle, and F. P. Imad, “Sensitivities of large multiple-loop control systems,” *IEEE Trans. Autom. Control*, vol. AC-10, pp. 308–315, 1965.
[17] *“DSA Tools Model Manual,”* ver. 7.1, Powertech Labs, Inc.

Durga Gautam (S’08) received the B.E. degree in electrical engineering from Institute of Engineering, Pulchowk Campus, Nepal, in 2002 and the M.E degree in electrical power system management from the Asian Institute of Technology, Pathumthani, Thailand, in 2005. Presently, she is pursuing the Ph.D. degree at Arizona State University, Tempe.

Her current research interests include power system dynamics, large scale integration of wind power, and distributed generation.

Vijay Vittal (S’78–F’97) received the B.E. degree in electrical engineering from the B.M.S. College of Engineering, Bangalore, India, in 1977, the M.Tech. degree from the Indian Institute of Technology, Kanpur, India, in 1979, and the Ph.D. degree from Iowa State University, Ames, in 1982.

He is currently the Director of the Power Systems Engineering Research Center (PSERC) and is the Ira A. Fulton Chair Professor in the Department of Electrical Engineering at Arizona State University, Tempe.

Dr. Vittal is a member of the National Academy of Engineering.

Terry Harbour (M’93) received the B.S.E.E. degree from Iowa State University, Ames, in 1993 and the M.B.A. degree from Kaplan University in 2007.

He is a Senior Utility Engineer employed in the utility industry.